

INSPO: Writing with Crowds Alongside AI

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Abstract

The use of artificial intelligence (AI) to support creative writing has bloomed in recent years. However, it is less well understood how AI compares to on-demand human support. We explored how writers interact with both AI and crowd worker writing assistants in creative writing. We replicated the interface of the prior crowd-writing system, HETEROGLOSSIA, and developed INSPO, a text editor allowing users to request suggestions from AI models and crowd workers. In a one-week deployment study involving eight creative writers, we examined how often participants selected crowd workers when fluent AI text generators were also available. Findings showed a consistent decline in crowd worker usage, with participants favoring AI due to its faster responses and more consistent quality. We conclude by discussing how crowd-writing systems, in the large language model (LLM) era, can shift to fostering LLM-human collaboration.

CCS Concepts

• **Computing methodologies** → **Natural language generation**;
• **Human-centered computing** → *User studies*; • **Information systems** → *Crowdsourcing*.

Keywords

Language Generation, Ideation, Crowdsourcing, Writing Support, Creative Writing

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1 Introduction and Backgrounds

Large language models (LLMs) are currently at the forefront of writing assistance research in HCI [25]. Their ability to generate fluent text and follow user instructions has sparked numerous projects in recent years, exploring how these capabilities might impact or support users' writing. Looking at the broader context of HCI, this is not the first wave of research focused on leveraging technology to assist writing. Before computers could fluently generate text in response to open-ended prompts, researchers were already pushing the boundaries of what writing assistants could achieve [16]. A notable example is the development of "crowd writing" systems [12]. These systems utilized online crowd workers, recruited programmatically from microtask platforms like Amazon Mechanical Turk, to perform tasks similar to those now handled by LLMs: producing text based on provided information and instructions. Such crowd-writing systems allowed researchers to study how users interacted with writing assistants that provided capabilities beyond the automated technology available at the time. For instance, SOYLENT explored how users can write using an editor that exhibited human-like intelligence [5]; KNOWLEDGE ACCELERATOR divided complex writing tasks into smaller pieces, which could be completed individually, and then reassembled into a cohesive final product [17, 23]; HETEROGLOSSIA explored how writers can work with story ideas generated from different characters' perspectives [19]. Like many crowd-powered systems, crowd-writing systems often share the vision of the "human computation" paradigm, as characterized by Quinn and Bederson: the functions operated by human workers would be eventually "solvable by computers." The use of crowd workers (temporarily) bridged the socio-technical gap between users' needs and system capabilities [1], enabling researchers to study user interaction and design challenges in anticipation of future technological advancements. The emergence of capable LLMs centers the question of *should we stop building crowd-writing systems?*

In this paper, we focus on creative writing, particularly story writing. Our key stance is that **humans generally are still much better at creative writing than the best LLMs**. Studies have shown that human-written stories are more creative [4, 36] and generally of higher quality [9], LLMs struggle with comedy writing [27], and

humans are also more effective at editing story drafts [10]. Therefore, in theory, crowd workers should still be capable of providing higher-quality support for story writers than LLMs, as humans are generally better at writing and editing stories. However, the reality is more complex. When considering literature comparing stories written particularly by online crowd workers against that of LLMs, findings are mixed: some studies suggest that crowdsourced stories offer more imaginative scenarios, settings, and rhetoric [3], while others show no differences [29]. Furthermore, despite LLMs generally not matching human ability in creative writing, certain steps of the writing process, such as overcoming writer's block or maintaining consistency [24], have shown to benefit from computational assistance [11]. It would be unwise to exclude AI from writing systems simply because it is not yet on par with human capabilities. As LLMs continue to develop, how crowd-writing systems can evolve to integrate computational support while preserving the advantages of human involvement is becoming an increasingly important research area.

As a step toward understanding AI's impact on crowd-writing systems, this paper examines writers' choices and interactions when presented with the option of using either crowd workers or AI as writing assistants. Specifically, we replicated the interface and in-situ suggestion delivery design of a prior crowd writing system, HETEROGLOSSIA [19], and developed **INSPO**, a text editor that allows writers to request suggestions from both AI models and online crowd workers (Figure 1). Writers using INSPO can select a portion of their draft and request suggestions from four different sources: **(i) crowd workers (Crowd)**, **(ii) a story plot generation model (Story Plot Generator)**, **(iii) GPT-3 for generating follow-up story arcs (GPT3-Plot)**, and **(iv) GPT-3 for continuing the given story (GPT3-Continuation)**. Through a one-week deployment study involving eight creative writers, we explored how often participants selected the crowd option when fluent text generators were also available (RQ1). Additionally, by analyzing their writing logs and user behavior data within INSPO, we examined how the suggestions generated by crowds and AIs affected users' writing quantity (RQ2) and content (RQ3), respectively.

2 INSPO System

Figure 1 shows INSPO's user workflow and interface, which follow the design of HETEROGLOSSIA by displaying suggestions beside the text editing panel and using highlighted colors to associate suggestions with the corresponding selected input text spans. The text editor (Figure 1A) allows users to write stories and request suggestions by selecting a text snippet and clicking the "Get Inspired" button. This opens the Task Creation Panel (Step 2 in Figure 1), where users can choose a generation method and set task-specific parameters like additional instructions or the number of suggestions. Upon clicking the "Create" button, INSPO generates suggestions and displays them in the Sidebar and Suggestion Panel. The Sidebar (Figure 1C) lists all suggestion tasks for the document, showing the task type and a brief snippet of the corresponding draft, and clicking on a task brings up the suggestion's content in the Suggestion Panel and links it to the relevant text snippet. The Suggestion Panel (Figure 1B) shows the generated suggestions and allows users to request more with the same settings by clicking the

"Get Another Plot" button. Each generated outcome is displayed in separate tabs (e.g., "Plot-1" and "Plot-2"), allowing INSPO to track which suggestions have been read. The text spans used for generating suggestions are highlighted in light blue (Figure 1A); and when a user hovers over a suggestion in the Sidebar (Figure 1C), the corresponding text snippet is highlighted in yellow (Step 1 in Figure 1).

Four Ways of Generating Suggestions. Writers using INSPO can select a portion of their draft and request suggestions from one of the following four different sources: **(i) crowd workers (Crowd)**, **(ii) a story plot generation model (Story Plot Generator)**, **(iii) GPT-3 that generates follow-up story arcs (GPT3-Plot)**, and **(iv) GPT-3 that completes the given story (GPT3-Continuation)**. These are called suggestion "task types", which we detail as follows: **(i) Crowd:** The Crowd task allows the user to post tasks on Amazon Mechanical Turk to ask crowd workers to provide plot ideas that continue the selected story snippet. Users can specify the number of ideas (the default is three) and write additional instructions for the workers. We adapted the code from HETEROGLOSSIA [19] for the implementation. **(ii) Story Plot Generator:** The Story Plot Generator option generates a few sentences describing possible next events in the story based on the selected snippet. This method predicts likely future events in a story based on its past [20]. We adapted Huang et al.'s data and code for the implementation. INSPO offered two models for users to select: one generates a *near future* plot, summarizing what would happen in the next 20 sentences; the other generates a *far future* plot, summarizing what would happen in the next 100 sentences. **(iii) GPT3-Plot:** We asked GPT-3 [7] to produce a follow-up story arc (see Appendix A for the prompt). **(iv) GPT3-Continuation:** This option continues the story based on the selected content by running GPT-3's text completion function (text-davinci-003).

3 User Study

We developed a proof-of-concept system, INSPO, and conducted a deployment study to examine how writers interact with the system when given the option to use either crowd workers or AI. The study combined quantitative and qualitative approaches: user behavior was analyzed through detailed log data, and these findings were complemented by qualitative insights gathered through interviews with participants. Table 1 overviews all the participants. Appendix B describes the participant selection criteria and recruitment details.

Two-Part User Study. In the study, we asked each participant to use INSPO to write short stories within the specified duration (e.g., 7 days.) The final outcome could be part of a longer story or a combination of short stories and did not need to be complete. Although we encouraged participants to meet a suggested word count (e.g., 2,000 words), it was not mandatory. We also encouraged them to use each of the four suggestion-generation functions daily. We recorded all user activities in INSPO, including keystrokes, mouse clicks, and timestamps for creating and reading inspiration tasks.

Our study had two parts, both of which used INSPO: a pilot study and the main study. **(1) Pilot Study (3 Days, 2 Participants):** The pilot study was 3-day-long, where the participants were encouraged to write 1,000 words. The compensation for each participant

Inspo's User Interface

(with the sidebar hidden)

User Interface
with Sidebar popped out



Figure 1: INSPO contains (A) a text editor, (B) a suggestion panel, and (C) a sidebar. Users can select any part of a working draft and send it to INSPO to request suggestions from either the text-generation model of their choice (including GPT-3) or online crowd workers.

was a \$100-dollar Amazon gift card. All other setups of the pilot study were identical to the main study. The pilot study led to final adjustments in INSPO, such as adding undo and redo functions and setting a minimum word count (50 words) for user-selected text. (2) **Main User Study (7 Days, 6 Participants):** The main study lasted 7 days, and participants were encouraged to write 2,000 words. We compensated each participant with a \$200 Amazon Gift Card. In both studies, on their first day using INSPO, participants completed

a self-serve onboarding tutorial. After each study, we conducted a 30 to 60-minute one-on-one semi-structured interview with each participant to gather feedback on INSPO. These interviews were audio-recorded for further analysis.

4 Analyses and Findings

We employed a mixed-methods approach, combining quantitative and qualitative analyses to address our three research questions

Part.	Study	Writing Exp.	Publishing Exp.	Days	#Word	#Task				#Suggestion			
						C	S	GP	GC	C	S	GP	GC
P001	Pilot	≤ 1 Year	Online forums, blogs, websites	3	1,376	<u>1</u>	<u>3</u>	<u>3</u>	<u>3</u>	1	3	<u>6</u>	<u>3</u>
P002		5-10 Years	Online forums, blogs, websites	3	1,352	3	<u>4</u>	2	<u>5</u>	5	<u>6</u>	2	<u>8</u>
P003	Main	5-10 Years	Published books	7	2,376	<u>6</u>	<u>6</u>	<u>6</u>	<u>12</u>	<u>6</u>	<u>6</u>	<u>6</u>	<u>12</u>
P004		≥ 10 Years	Published books	7	10,289	<u>6</u>	<u>11</u>	<u>6</u>	<u>6</u>	<u>12</u>	<u>11</u>	7	7
P005		5-10 Years	Online forums, blogs, websites	7	1,773	1	<u>2</u>	1	<u>5</u>	<u>2</u>	<u>2</u>	<u>2</u>	<u>9</u>
P006		3-5 Years	Online forums, blogs, websites	7	2,317	2	4	7	<u>5</u>	2	6	<u>9</u>	<u>8</u>
P007		≥ 10 Years	Online forums, blogs, websites	7	2,715	5	<u>6</u>	5	<u>15</u>	<u>11</u>	10	8	<u>16</u>
P008		≥ 10 Years	Online forums, blogs, websites	7	2,759	6	<u>9</u>	7	<u>10</u>	<u>16</u>	<u>19</u>	14	<u>19</u>
Total	-	-	-	-	-	30	<u>45</u>	37	<u>61</u>	55	<u>63</u>	54	<u>82</u>

Table 1: Participant information and statistics. Part.: Participant, Writing Exp.: Writing Experience, Publishing Exp.: Publishing Experience, C: Crowd, S: Story Plot Generator, GP: GPT3-Plot, GC: GPT3-Continuation. P001 and P002 joined the pilot study; the rest joined the formal user study. The highest and second highest usage are highlighted.

Task Type		C	S	GP	GC
#Samples		30	45	37	61
System	25%	1,753	7	7	9
Latency	50%	3,449	8	8	9
	75%	8,160	10	9	10
#Samples		19	36	32	51
Reading	25%	21,881	15	8	4
Latency	50%	34,429	115	20	13
	75%	73,652	538	74	84
Not Reading		36.67%	20.00%	13.51%	16.39%

Table 2: System and reading latency (in seconds) for four different tasks (RQ1). C: Crowd, S: Story Plot Generator, GP: GPT3-Plot, and GC: GPT3-Continuation. For Crowd, half took up to 60 minutes for an initial response, and users typically read the response hours later. A higher rate of not reading is also observed.

(RQs). To gather supportive evidence, we examined user behavioral and writing logs. Additionally, we transcribed all interviews and conducted analytic induction, a method that blends deductive and inductive approaches [33, 39]. Our initial analysis of the interview data yielded 287 unique quotes or phrases from participants. We developed a codebook and applied it to categorize 67 distinct excerpts. These excerpts were then systematically grouped to address our three RQs. Note that as the pilot study proved successful, we merged its data with the main study for analysis.

4.1 RQ1: How frequently is each suggestion generation method used?

Table 1 shows the usage statistics of the whole study. P004 incorporated their original works into the writing task, resulting in an extensive final draft of 10,287 words. In contrast, P008 produced three shorter drafts, with word counts of 1,198, 1,288, and 272, respectively.

4.1.1 GPT3-Continuation was the most popular, while Crowd usage declined after the initial phase. Table 1 shows that the GPT3-Continuation function was the most popular, with 74 tasks created and 95 suggestions generated. Six participants used GPT3-Continuation

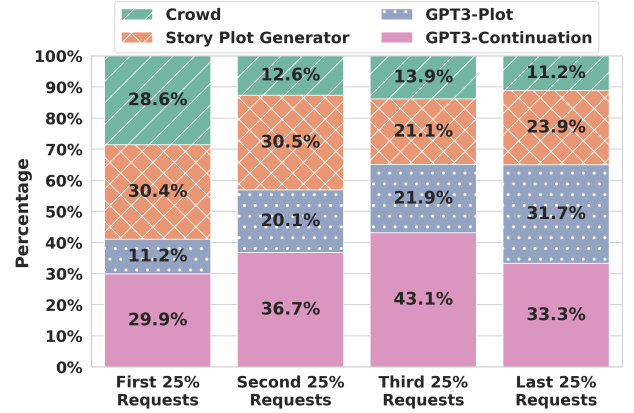


Figure 2: Usage distribution of four functions across study phases, highlighting a decline in the Crowd function from 28.6% initially to 11%-13% in subsequent phases (RQ1).

most frequently. We further analyzed the percentage of each function's use within the first, second, third, and last 25% of participants' requests to INSPO. For instance, if a participant made 100 requests, we grouped them into four sequential batches of 25 requests each. Figure 2 shows that participants reduced their use of the Crowd option after the initial phase of the study. Initially, the crowd function accounted for 28.6% of requests, but this dropped to 11%-13% in later batches. From interviews with participants, we found that **all (8 out of 8) participants unanimously preferred GPT3-Continuation**. They appreciated the immediate responses provided by GPT3-Continuation, which offered "instant gratification" (P002) and helped them overcome writer's block by generating fresh ideas. Some participants mentioned that GPT3-Continuation sometimes proposed alternative leads for completion, sparking new directions in their writing. P005 noted, "(GPT3-Continuation) actually worked and worked very perfectly." On the other hand, participants valued the human touch and originality in the Crowd inputs, expressing appreciation for the personal touch and shared understanding humans brought to their writing. For example, P007 mentioned that "it made logical sense, any of the ideas or the continued story that was

given. I think it (Crowd) was the most helpful overall.” Participants also enjoyed the ability to ask more directed questions and explore different alternatives for their writing. Nevertheless, participants faced significant issues with inconsistent and delayed responses from the Crowd function. As P003 noted, “the timeframe was just... it’s hard to not compare it to the other ones, the other ones were just so much faster. And so that was like the big drawback.”

4.1.2 Crowd had a much longer system and reading latency, with one-third of responses going unread. From interviews with participants, we learned that latency significantly influenced their choices. To analyze this, we calculated two types of latency for each function in INSPO: (i) system latency and (ii) reading latency. System latency is the time from when a user creates a request to when the system first responds. Reading latency measures the time from receiving the first response to when the user reads it. Table 2 shows the results. Most AI functions delivered responses within 10 seconds; users typically read these suggestions immediately. In contrast, the Crowd function could take up to an hour for an initial response, and users often delayed reading these for several hours. Moreover, participants did not read 36.67% of the suggestions produced by the Crowd function.

4.2 RQ2: How does each method of suggestion affect word count?

RQ2 examines whether INSPO suggestions impact word count, a rough yet commonly used measure of writing progress. We analyzed participants’ writing logs by first identifying active user sessions using a cluster-based approach. Within each session, we used NLTK [6] to calculate the document’s word count when the suggestion panel was accessed and again 5 (or 3) minutes later. Detailed analysis steps are described in Appendix C. Across the study, we observed 171 working sessions, with a mean duration of 609 seconds (SD=973) and a median duration of 257 seconds.

To meaningfully interpret these word count changes, we needed to compare them against typical writing productivity without using INSPO suggestions. This comparison would allow us to determine whether INSPO actually influenced writing output beyond normal productivity patterns. Therefore, we introduced a **Baseline** to capture writing behavior without INSPO suggestions. The Baseline was estimated by running the following simulation 1,000 times: We randomly selected a working session, with the probability of selection proportional to the session’s length. We then randomly picked a starting point within this session and measured the document changes 5 (or 3) minutes later. Figure 3 shows KDE plots comparing the four suggestion functions with the baseline.

4.2.1 GPT3-Continuation boosted word production the most. In Figure 3a, the distribution from GPT3-Continuation is flatter than all other options and the baseline. GPT3-Continuation’s main peak around 0 words—indicating no additional words produced—is also the lowest among all options. These suggest that GPT3-Continuation is the most effective in boosting word production. Story Plot Generator and GPT3-Plot, while not as effective as GPT3-Continuation, still made participants produce more words than the baseline. We performed the same analysis with a 3-minute window (Figure 3b), GPT3-Continuation remains superior. The Crowd function

shows the highest main peak at 0 words, suggesting participants often did not produce any words within 5 minutes of reading its suggestion. We hypothesize that this occurs because Crowd suggestions frequently relate to drafts written hours earlier, requiring users to spend time reacquainting themselves with the context before continuing their writing. Furthermore, a more pronounced peak at 0 words in a 3-minute analysis window (Figure 3b) indicates that participants were even less likely to incorporate Crowd suggestion within 3 minutes, and slightly more likely to do so within 5 minutes. Our interviews show more nuanced details on how different suggestion-generation methods influence participants’ writing process. The immediate availability of GPT3-Continuation suggestions appears to contribute to its popularity and effectiveness in boosting word production. In contrast, while the delayed nature of Crowd suggestions could be frustrating at times, some participants appreciated the depth it added to their writing process. One participant said, “Sometimes I needed to see how what I’d written had come across before moving on to the next part, and I was able to get a response from an actual person. Like, an actual reader, essentially, to see if what I was saying made sense to them or what they thought was happening.” (P007).

4.3 RQ3: How does each suggestion method influence writing content?

We compared sentences in the suggestions to the “newly-edited content” in the drafts. “Newly-edited content” includes (i) sentences that were not in the draft when the suggestion was accessed but appeared five minutes later, and (ii) sentences that were edited within these five minutes. For each suggestion-reading event (see RQ2 and Appendix C), we first calculated sentence-level similarities pairwise. For example, with a 3-sentence suggestion and a draft having 5-sentence newly-edited content, we calculated 15 (3×5) similarities. We then took the highest value among the 15 as the final similarity score, considering that even one closely matching sentence indicates a strong influence [34]. We applied three sentence-level similarity metrics: (i) **Normalized Edit Similarity**: $1 - \text{NormalizedEditDistance}$ [26, 38], where higher scores indicate higher similarity; (ii) **Semantic Similarity**: cosine similarity on semantic representations obtained from Sentence BERT [32]; and (iii) **Paraphrasing Detection Score**: an output probability from a paraphrasing detection model [28] to determine if a written sentence is a paraphrase of one of the suggestions.

Baseline: We simulated a baseline condition by comparing a random working draft and a random suggestion. We randomly selected a time from a working session (as defined in RQ2) and observed changes five minutes later. We also randomly sampled one suggestion from the same document and then applied the three similarity metrics to these sentences. The simulation was run 1,000 times. Finally, similar to RD2, we used KDE plots to visualize the influence levels across the four suggestion methods and the baseline condition. Figure 4 shows the results, in which a 1.0 similarity indicates complete integration of the “suggestion sentence” into the user’s draft, while a 0.0 similarity means no integration occurred.

4.3.1 GPT3-Continuation influenced the written content the most, with full-sentence adoption across all methods. Across all three charts in Figure 4, GPT3-Continuation’s curves are flatter and

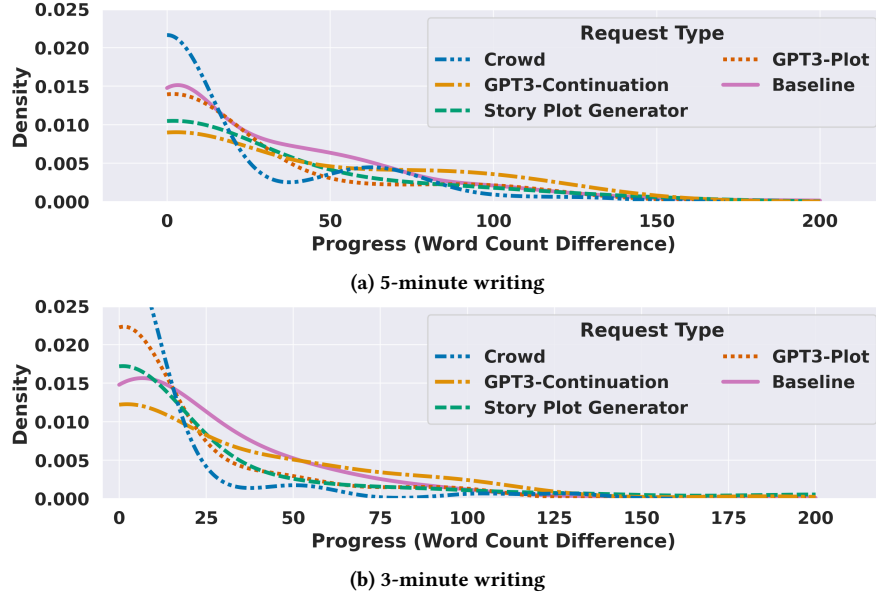


Figure 3: Distribution of the writing progress (RQ2). In (a), GPT3-Continuation shows a higher density near 100 words; Crowd comes with a small peak at 70 words. This peak reduces and shifts left in (b), indicating that users might need time to catch up with the prior information. (We set `common_norm` to false to ensure each distribution’s area sums to one. We also clipped values to $[0, 200]$.)

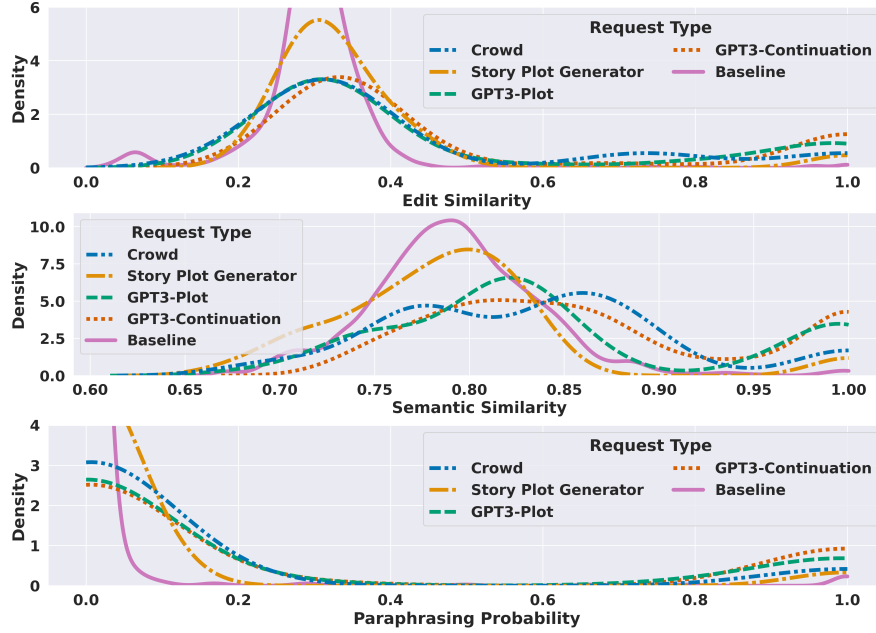


Figure 4: Distribution of the similarity metrics (RQ3). A 1.0 similarity means that the whole sentence is adopted. Overall, GPT3-Continuation has a higher rate of being adopted.

skewed to the right, indicating a higher likelihood of users producing texts similar to those generated by GPT3-Continuation. Additionally, at the high-similarity end (close to 1.0), GPT3-Continuation exhibits a greater probability density compared to other

options. This indicates a higher tendency for GPT3-Continuation sentences to be directly copied into participants’ drafts. Overall, GPT3-Continuation influences the participants’ writing the most.

Note that, at the high-similarity ends, all options exhibit a probability density greater than zero, indicating that full-sentence adoption occurs across all suggestion methods in INSPO.

From the interviews, participants confirmed that GPT3-Continuation often directly influenced their writing content, with some participants incorporating sentences or phrases directly from the AI-generated suggestions into their drafts. Meanwhile, participants had mixed feelings about using GPT3-Plot and Story Plot Generator functions. Some used GPT3-Plot to validate their storylines or explore alternative perspectives. P003 mentioned, *“the function was helpful for figuring out how to map out a plot. I didn’t really start with a plot in mind when I was writing this story.”* However, others found that GPT3-Plot and Story Plot Generator sometimes produced irrelevant suggestions. P007 noted, *“I stopped using the near [future option] after a couple of days because I wasn’t getting anything helpful from it.”* Participants also noted that the AI sometimes struggled with complex prompts and did not always align with their expectations, which could limit the influence of these suggestions on their writing content.

5 Discussion and Design Suggestions

As noted in the Introduction (Section 1), one key mission of crowd-powered systems is to bridge the social-technical gap between what users want and what computers can do [1], especially when technology is not yet advanced enough. Specifically, crowd workers perform tasks that computers cannot yet handle (such as generating fluent text), allowing researchers to study how users interact with these (future) capabilities in order to figure out the design, system architectures, and interactions for them. Furthermore, some crowd-powered systems can also serve as scaffolds for future automation by providing insights into how automation can gradually be integrated into human-operated, already-working systems and also by collecting data to train or improve AI models. To some extent, our findings suggest that, with LLMs now widely accessible, we no longer need to use crowd workers to simulate how (future) language technologies could support writing: the future has arrived; although imperfect, LLMs are here to stay.

Should we stop building crowd-writing systems? We do not think so. When one set of missions that crowd-writing systems were initially set to do fades away, another mission becomes more salient. Historically, one important mission of crowd-powered systems—besides filling the social-technical gap in research—has long been to explore how a collection of well-scoped human work can be assembled, coordinated, and integrated with one another, as well as with automated components and users, to tackle larger, complex, and challenging tasks [12, 16, 23]. Examples like CrowdForge [23], Knowledge Accelerator [17], and Mechanical Novel [22] have particularly focused on creating long, complex written products. **With LLMs widely available, understanding how to integrate these fluent text generators into complex, dynamic human collaborations is more relevant than ever.** Crowd-writing systems offer an ideal testbed for experimenting with such integrations: By programmatically recruiting large groups of online workers and using algorithmic workflows, these systems enable testing different strategies for distributing writing tasks between humans and AI. That being said, our study highlights one key challenge: without

thoughtful design, these systems can lead to *AI overreliance* [37], where users often ignore human suggestions in favor of AI, which is faster, always available, and of acceptable quality. While AI reliance has been studied in other contexts [8, 37], we offer two design suggestions specifically for creative writing based on our results: **First, account for differences between human and AI assistance.** In our study, human assistance differed significantly from AI. The most obvious disparity was their response speed; participants also noted greater variability in the quality of the crowd assistance, while mentioning that the crowd options sometimes offered a more “human touch.” However, our study followed the design of HETEROGLOSSIA, which delivered both AI and human assistance through an identical modality and interface. This proved ineffective. Future systems should be more intentional in emphasizing and accommodating these differences. **Second, design against automation bias.** Our study shows that when AI is presented as an (seemingly equal) option alongside crowdsourcing in writing, users tend to favor AI (RQ1). As a result, AI has more influence on the writing process: more words are generated based on AI suggestions (RQ2), and more of the content is shaped semantically by AI’s output (RQ3). This influence may not always be desirable, particularly in creative writing scenarios where humans often outperform AI [4]. To ensure users make thoughtful decisions between AI and crowd assistance when using future crowd-writing systems, our second design suggestion is to design against *automation bias*—the human tendency to over-rely on or accept machine-generated outputs [14, 15]. Designers of writing systems must be mindful of this bias and work to mitigate the potential negative impact of AI on users’ writing.

6 Conclusion and Future Work

This paper introduces INSPO, a text editor that allows creative writers to request inspiration from online crowd workers and three different AI models, including GPT-3 for continuing ongoing stories (GPT3-Continuation). A week-long user study with multiple creative writers showed that participants consistently decreased their use of crowds in favor of AIs throughout the study. Motivated by these findings, we discuss how the role of crowd-writing systems can evolve in the era of LLMs, focusing on fostering LLM-human collaboration, especially in complex collaborative settings. Future work will involve larger-scale studies, enhanced INSPO features, and the integration of more advanced and diverse LLMs.

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A Prompts Used in GPT3-Plot

We asked GPT-3 [7] to produce a follow-up story arc using the prompt:

*Given the previous story: [story-snippet].
Follow the instruction: [instruction] to describe the
follow-up story arc using 50 words.*

where [story-snippet] and [instruction] are the placeholder for the selected snippet and the instruction.

B Participant Recruitment and Criteria for Participant Selection

B.1 Pilot Study

Two participants were recruited via the authors’ networks for the pilot study.

B.2 Main Study (Section 3)

For the main user study, the participants were recruited via the authors’ networks, social media posts, and mailing lists in the authors’ institute. From the 37 individuals who expressed interest, 6 participants were chosen based on two primary criteria:

- First, preference was given to individuals with experience in publishing books, whether through traditional publishing houses or self-publishing platforms. Recognizing that most people in our recruitment pool had limited or no publishing experience, prioritizing published authors allowed us to include a broader range of writing expertise.
- Second, participants were selected based on their writing processes. This included distinguishing between those who outlined their stories before writing (outliners/architects), those who developed their stories organically while writing (discovery writers/brainstormers/gardeners), and those who used a hybrid approach [31]. This diversity ensured a representative sample of writing practices in the participant pool.

C Detailed Procedure of Analyzing Writing Logs (RQ2 & RQ3)

To perform a comprehensive analysis of writing logs, our approach involves several key steps:

Step 1: Reconstructing Texts at Any Timestamp. We reconstructed the participants’ writing at time T by collecting all text-change events occurring before time T . Each event includes a keystroke and a cursor offset. This set of text-change events was then rendered into a Delta document using `quill-delta` [35]. Next, we converted the Delta document to HTML using `Quill-delta-python` [13] and extracted the text with `html2text` [2].

Step 2: Recreating Drafts Before and After Reading suggestions. We aim to measure the impact of suggestion on writing behavior, specifically after the writer reads the suggestion. We extracted “suggestion-reading events” where participants interacted with the suggestion panel’s tab by clicking and opening it. We then reconstructed and compared the state of the draft at two specific times: the moment of accessing the suggestion panel (time of the suggestion-reading event) and 300 seconds (5 minutes) thereafter. We also prepared a comparison for 180 seconds (3 minutes).

Step 3: Defining Working Sessions. We aim to simulate “baseline” writing behavior, helping us distinguish between short pauses for thinking and actual breaks from the task. We followed a prior work [18] to identify these sessions. We first compiled all activity logs, including text changes and suggestion-reading events, and divided them into working sessions using an “inactivity threshold.” After testing thresholds ranging from 1 to 20 minutes, we recorded

the number of sessions for each threshold. Using the elbow rule [21], we identified 240 seconds as the optimal threshold, which gave us 171 working sessions with a mean duration of 609 seconds ($SD=973$) and a median duration of 257 seconds.